

Extraction and Analysis of American Universities' Privacy Policies and DNSMPI Links

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1 Introduction

Modern technology has led to the processing and collection of data at an unprecedented scale [2]. This comes with both great opportunity and risk [2]. One potential risk is the sale and compromise of personal privacy information [2].

The legal concept of privacy in the context of personal information allows for the consensual transaction of a consumer's personal information in exchange for access to a good or service [2]. However, it is a violation of the consumer's privacy and inherent rights protected under privacy law when personal information is sold, shared, or compromised without the consent of the consumer [2]. Increasingly, the control over consumers' personal information has shifted to avaricious third-parties, who may not have the best interest of the consumers in mind and may even be completely unknown to the consumer [2].

Due to concern over protecting personal information, numerous laws tailored to this issue have been passed. One of the most comprehensive yet complicated laws [3], the California Consumer Privacy Act (CCPA), later modified as the California Privacy Rights Act (CPRA), has intended to hold certain for-profit companies accountable by requiring a Do Not Sell My Personal Information (DNSMPI) link on their homepages to allow consumers to opt-out of disclosing their personal information [7]. However, laws addressing the disclosure of personal information are relatively recent, and it is still unclear how effective and enforceable they are, especially regarding entities outside of California [3].

This paper applies the topic of privacy regulation and enforcement in the context of one of the largest and most diverse industries in the US: education. In 2022, an estimated 15.9 million students enrolled in American universities [10]. Despite this, no major research has investigated American universities' privacy policies. Hence, this project attempts to bridge the gap in existing research between privacy

policies and universities by performing an extensive search, collection, visualization, analysis, and discussion of American universities' privacy policies.

2 Research Questions

2.1 Research Question 1 (RQ1)

Research question one assesses the level of cultural or regulatory impact the CCPA (modified as the CPRA) has had on universities' voluntary adoption of DNSMPI links. Specifically, the study investigates if California's laws and emphasis on data privacy have created a local "industry standard" of DNSMPI link adoption and data transparency.

Question

Is there a disproportionately higher voluntary adoption rate of DNSMPI links by California universities than universities outside of California?

2.2 Research Question 2 (RQ2)

Given the relatively recent enactment of the CCPA (modified as the CPRA) and its application to some for-profit universities outside of the state of California, measuring the success and scope of DNSMPI link regulations is beneficial. In particular, seeking to compare the estimated compliance rate among American universities with the compliance rate of the most visited websites explored by Van Nortwick et al. in 2022.

Question

What is the estimated compliance rate with DNSMPI link requirements among American universities, and which universities are uncooperative?

2.3 Research Question 3 (RQ3)

To better analyze the diversity of the industry, American universities are grouped into three distinct categories: public, private non-profit, and private for-profit. These groups were chosen because they represent differing business models, funding sources, legal obligations, stakeholder accountability, and institutional goals. Such variations raise an intriguing question: do these differences influence how transparent or obfuscatory universities are in their privacy policies? Specifically, do some types of universities craft policies that are more clear and accessible than other types? Do some types lean toward vagueness, potentially to confuse, mislead, or manipulate readers? Lastly, what can the privacy policies describe about their business models?

Privacy policies are not just legal documents; they can reflect institutional priorities and strategies. For instance, a policy with positive sentiment and optimistic language may foster trust, while an excessively long policy with complex wording could hinder comprehension. By analyzing metrics such as word length, reading level, keywords, and sentiment, the question aims to uncover how these universities communicate with their audiences. Are public universities more transparent due to their accountability to taxpayers and government funding? Do for-profit institutions, driven by business objectives, prioritize ambiguity to obscure certain practices? The research explores these questions to show how privacy policies reflect institutional goals.

Question

How do privacy policies' readability (reading level, word length, sentiment) and content (policy acknowledgments) differ among public, private non-profit, and private for-profit universities?

2.4 Research Question 4 (RQ4)

This study has identified three main classes of universities to compare against each other for the aforementioned reasons of different business models, funding sources, legal obligations, accountability to stakeholders, and institutional goals. However, the most significant privacy policy content may reveal more nuanced motives and underlying groupings of universities beyond university class. Hence, this study seeks to determine the strength of the correlation between university class and word choice, uncover universities' legal perspectives and priorities, and identify any potentially useful groupings or clusters of universities beyond class for further research.

Question

What groupings or clusters of universities can be identified based on the significant wording in their privacy policies? To what extent do these clusters align with traditional university classifications (public, private non-profit, private for-profit)? What does significant wording reveal about the legal prioritization of universities' privacy policies for each cluster?

3 Methodology

3.1 Methodology for the Crawler

3.1.1 Crawler Description

To scrape information from the selected sample data, this project employs a combination of automated browser interaction, HTML parsing, and content extraction to locate and access privacy-related pages from websites. The asynchronous Python script, *crawler.py*, is designed to scrape websites for privacy

policies and DNSMPI (Do Not Sell My Personal Information) links and format the information in JSON objects for future analysis.

3.1.2 Crawler Description

Input preparation begins with a CSV file containing two main columns: homepage URLs of websites to scrape and their corresponding classifications, which label the research context as “Public,” “Private for Profit,” or “Private Non-Profit.” These classifications provide a framework for analysis.

The automated scraping process uses Python and the Playwright library to simulate headless browser actions. Relevant pages are located by searching for keywords such as “privacy policy,” “privacy notice,” and lastly “privacy” for privacy-related content, and terms like “Do Not Sell,” “California Privacy Rights,” “Do Not Share,” and “DNSMPI” for DNSMPI content. To optimize efficiency, the scraper uses Python’s Asyncio library to operate asynchronously, with a semaphore limiting resource usage. Error handling is incorporated to log data collection timeouts, navigation errors, and flagging issues.

Content extraction focuses on retrieving the main content from the identified privacy policy and DNSMPI pages using the BeautifulSoup4 Python library. The methodology finds HTML tags, such as <main> or <article>, to isolate relevant text. For completeness, content from multiple pages is consolidated when necessary. The structured output is saved in a JSON file, capturing essential details such as the website URL and classification (as provided in the input CSV), the status of whether a privacy policy link was found, the link itself, if available, and the extracted privacy policy text. Similarly, the DNSMPI link and its extracted text are included if found. Any errors encountered during the scraping process are logged under an “Errors” field for further review.

3.1.3 Crawler Results

Overall, the crawler obtained the selected sample universities' privacy policy text of 90.7% (137/151). Out of the fourteen failures, the crawler obtained the privacy policy link for three of them. The remaining fourteen policies were added to the dataset manually. Moreover, the crawler was manually verified for the DNSMPI link, and it found the crawler successfully made this distinction 100% of the time. Of the eight DNSMPI links, the crawler successfully retrieved the text of five of them. Consequently, the content was added for the three remaining DNSMPI links.

3.1.4 Crawler Limitations

It is important to note that the crawler may not always be effective at locating necessary links due to heuristic matching, HTML structural limitations, and website timeout issues. Timeout issues often arise when websites contain excessively long HTML bodies, which can exceed Playwright's timeout thresholds.

3.2 Methodology for Privacy Policy Data Extraction

3.2.1 Extraction Description

The *extract_data.py* script processes JSON data containing university privacy policies and analyzes them to extract insights on legal mentions, text readability, sentiment, and word count. It identifies references to CCPA/CPRA, FERPA, and GDPR, calculates reading ease using the *Flesch Reading Ease Formula*, and determines sentiment polarity using *TextBlob*. The results are compiled into a structured dataset and saved as CSV and Excel files for further analysis.

3.2.2 Extraction Implementation

First, the script locates references to the following privacy laws: California Consumer Privacy Act (CCPA), California Privacy Rights Act (CPRA), Family Educational Rights and Privacy Act (FERPA), and General Data Protection Regulation (GDPR) through simple keyword matching. Secondly, the script analyzes word count, scores readability using the *Flesch Reading Ease Formula* from the *TextStat* library (ranging from 0 for very confusing to 100 for very easy), and determines sentiment using the *TextBlob* library, which returns a polarity score between -1 (negative) and 1 (positive). Additionally, the script checks for a DNSMPI link and flags its existence.

The results are compiled into a structured Pandas DataFrame and saved in a CSV file and an Excel workbook. The output includes fields such as the university link, type of university, legal mentions, DNSMPI presence, word count, readability score, and sentiment score, offering a comprehensive data view.

3.2.3 Extraction Limitations

Legal mentions are identified through keyword matching, which may result in false positives or negatives if synonyms or context-specific phrasing are used. Furthermore, readability scoring relies on the *Flesch Reading Ease Formula* from the *TextStat* library, which may not account for non-standard text structures. Similarly, sentiment analysis uses *TextBlob*, based on general-purpose language models, which might inaccurately interpret domain-specific language found in policy texts. Furthermore, the tool assumes all policies are in English and does not handle alternate languages.

3.3 Methodology for Visualizing the Extracted Data

3.3.1 Visualization Description

To visualize table data consisting of word length, keyword acknowledgment, reading, and sentiment scores, this study created two scripts called *visualize.py* and *analyze_chart_q3.py*. Using the csv table created from *extract_data.py*, these scripts create and save bar graphs, bar plots, and scatter plots. These

visuals help show patterns, comparisons, and averages among the three kinds of universities: public, private non-profit, and private-for-profit.

3.3.2 Implementation Details

The *visualize.py* script uses Python libraries such as *Pandas*, *Matplotlib*, and *Seaborn* to process and visualize data from a CSV file. The CSV file is assumed to follow the structure in the *Data Analysis Table* below. The script begins by loading the CSV file into a *Pandas DataFrame*. Then, insights are taken by using *Pandas* methods such as *value_counts()* to count the mentioning of privacy policies (CCPA, FERPA, etc.) and *groupby()* to compute the averages for metrics like word length and reading level. These results are then visualized by university type through *Matplotlib* and *Seaborn*.

Like *visualize.py*, the *analyze_chart_q3.py* script uses *Pandas* and *Matplotlib*, assumes the same structured CSV file, and loads the CSV data into a *Pandas DataFrame*. Using the *Pandas groupby()* method, the data is grouped by university type (public, private non-profit, and private for-profit) to compute averages for metrics such as word length, reading level, and sentiment index. Next, proportions of acknowledgment of privacy standards like CCPA, FERPA, and GDPR are calculated by applying the *mean()* method to the grouped data. These results are then visualized into a bar graph and a text file.

University link	Kind of university	CCPA or CPRA	FERPA	GDPR	DNSMPI	Word length	Reading level	Sentiment index
https://westcoastuniversity.edu/	Private For Profit	0	0	0	0	2276	28.67	0.100970206
https://plazacollege.edu/	Private For Profit	0	0	0	0	564	37.13	0.029392447
https://americancareercollege.edu/	Private For Profit	0	0	0	0	999	15.44	0.043617326
https://an.edu/	Private For Profit	0	0	0	0	893	32.94	0.06440281
https://www.phoenix.edu/	Private For Profit	1	1	0	1	10880	33.24	0.078743181
https://www.ecpi.edu/	Private For Profit	0	0	1	0	1464	38.55	0.056496599

Data Analysis Table- Shows six entries from CSV table used in visualization scripts

3.4 Methodology for Determining the Applicability of *Do Not Sell My Personal Information* Requirements

Do Not Sell My Personal Information (DNSMPI) requirements vary according to the kind of university. Non-profit businesses and government entities, such as public and private non-profit universities, are automatically exempt from implementing a DNSMPI link [7]. A for-profit entity like a private for-profit university is only applicable to DNSMPI obligations if it has an annual gross revenue of at least \$25 million per year, or it sells the personal information of at least 50,000 Californians per year, or it obtains a majority of its revenue from the sale of Californians' personal information [7].

Determining the gross revenue for a for-profit university was simple when provided with the expected cost of attendance and the number of enrolled students in a given year. However, the other two criteria were either impractical or unnecessary to calculate.

The people for whom for-profit universities are likely to have personal information are current and former students, employees, and university applicants. Given that the population of California relative to the national population is about 12% and assuming a uniform distribution of people for whom they have personal information [7], the for-profit universities would have to sell roughly 417,000 Americans' personal information in a given year. Furthermore, under the Family Educational Rights and Privacy Act (FERPA), if a student were to receive federal financial aid, then a for-profit university would be barred from selling such a student's personal information [8]. Many for-profit universities that were investigated that did not exceed 25 million in gross revenue had small student populations (under 1,000 students). Hence, it would be highly unlikely for any for-profit university with less than 25 million dollars in gross revenue to have access to and sell that many people's personal information.

Moreover, the business model for private for-profit universities is that they make most of their revenue from tuition and not from the sale of personal information, which was consistent with the reviewed data. Hence, gross revenue requirements should be considered when determining if a university is obligated to have a DNSMPI link.

In addition to DNSMPI data, universities, university campuses, or university headquarters located in California were also distinguished. This data was useful in determining the voluntary rate of DNSMPI link adoption inside and outside of California and identifying any defiant universities with respect to DNSMPI obligations.

All the manually compiled DNSMPI data is available in *A5/data/dnsmpi_data.xlsx* and *A5/data/dnsmpi_requirements.xlsx* in the repository.

3.5 Methodology for the Statistical Analysis of the Collected Privacy Policy Data

3.5.1 Statistical Tests

The *data_stats.py* script utilized the *statsmodels* module for both t and z tests. A two-sample t-test was conducted for the following metrics: word length, reading level, and sentiment index. A two-sample z-test was performed for tests involving the proportion of DNSMPI adoption and the individual proportions of privacy policies that mention the following terms: CPPA or CPRA, FERPA, and GDPR. All of the tests were defined at the .05 significance level.

3.5.2 Data retrieval

Data_stats.py utilized the *pandas* module for retrieving relevant data from *corrected_output.json*, *dnsmpi_data.xlsx*, and *dnsmpi_requirements.xlsx*.

3.5.3 RQ1

Initially, the *data_stats.py* used the *pandas* module to compile a list of voluntary DNSMPI adoptees that had a DNSMPI link on their homepage but were not required to have one. Then, *data_stats.py* tested for a significant difference in the proportion of voluntary DNSMPI link adoption between universities inside and outside of California. Additionally, the script calculated the voluntary link adoption rate among California universities, non-California universities, and all American universities investigated. This information was stored in *A5/analysis/dnsmpi_analysis.txt*.

3.5.4 RQ2

First, *data_stats.py* gathered the universities required to have the DNSMPI link. Then, the script estimated the DNSMPI link compliance rate among American universities by finding the proportion of universities required to have the DNSMPI link that did have it. Moreover, *data_stats.py* listed the universities that were not compliant with their DNSMPI obligations. This information can be found in *A5/analysis/dnsmpi_analysis.txt*.

3.5.5 RQ3

data_stats.py performed pairwise tests via *statsmodels* among university classes with respect to their privacy policies' mean word length, reading level, sentiment index, and the proportion of CCPA or CPRA, FERPA, and GDPR acknowledgment. See *A5/analysis/pairwise_tests.txt* for the data.

3.5.6 Descriptive Statistics

The *statistics* module was used to calculate descriptive statistics for the proportion of the mentioning of certain laws (CCPA or CPRA, FERPA, GDPR), word length, reading level, sentiment index, and the proportion of DNSMPI adoption for each university class and the total dataset. This data is located in *A5/analysis/descriptive_stats.txt*.

3.6 Methodology for the Extraction and Analysis of Significant Wording (RQ4)

This study defines a significant word as a word with a high Term Frequency-Inverse Document Frequency (TF-IDF) index. The term frequency $TF(w,d)$ is the number of times a word w appears in a document d over the number of words in that document [1]. The document frequency $DF(w)$ is the number of documents in which the word w appears [1]. Let D be the total number of documents. The Inverse Document Frequency is defined as:
 $idf(w) = \log(D/df(w))$ [1]

Lastly, the TF-IDF index for a particular word w and specific document d is calculated as:

$$TF\text{-}IDF(w,d) = TF(w,d) * IDF(w) [1]$$

This score prioritizes words with a high word count but penalizes words appearing in multiple documents [1]. In this way, frequency and uniqueness are counterbalanced.

cluster.py gathered the privacy policies from *corrected_output.json* and removed stop-words (e.g., a, the, is, etc.), words that appeared in more than half of the policies (e.g., privacy, data, sell, etc.), and words that appeared in just one document (e.g., Yale, Harvard, Stanford, etc.). The remaining words to analyze are considered potentially significant words. Next, each document was vectorized via the *TfidfVectorizer* module. This added the TF-IDF index of each potentially significant word as an entry in a vector corresponding to each document.

A *k-means* approach with the *sklearn* module was utilized to cluster documents with similar significant wording. For k groupings or clusters, the *k-means* approach randomly selects k vectors within the dataset to serve as the initial clusters [5]. A document cluster has a centroid that takes the mean value across all vector entries in that cluster [5]. A vector is added to a cluster if that vector is the closest to that cluster's centroid [5]. This iteratively continues until all vectors have been placed into k distinct clusters [5]. The process was run 100,000 times, and the version with the least total variation between the vectors and its clusters' centroid was used.

The optimal value of k is determined by looking at the graph *optimal_k.png* with varying k 's between 1 and 10 and selecting the one at the “elbow” in the graph such that the marginal decrease in the sum of the distances from each vector to its cluster's centroid is relatively low for all values of k greater than the optimal value.

Once the documents were clustered, the centroid of each cluster was located. The entries were retrieved with the highest values corresponding with the most significant words with a maximal TF-IDF index. This approach enabled the determination of the number of clusters, the universities' privacy policies within each cluster, and the most significant words in each cluster.

See A5/analysis/README.md in the repository for more detailed information on extracting and analyzing significant wording.

4 Results

All the developed code and the full results are available in the following repository:

<https://research-git.uiowa.edu/bhourcade/cs3640>

4.1 RQ1

Of the 83 non-California universities not under DNSMPI obligations, none of them implemented a DNSMPI link. In comparison, of the potential forty California universities, just one voluntarily adopted a

DNSMPI link. The p-value for the test for the voluntary adoption rate of DNSMPI links between California and non-California universities was .3118, which concludes that there is not enough evidence that the voluntary DNSMPI adoption rate is higher in California. See A5/analysis/dnsmpi_analysis.txt in the repository for the full results.

4.1.1 Insight and Implications for RQ1

This result shows that implementing the DNSMPI link has conclusively not become an industry standard among universities. Because the CPPA came into force on January 1st, 2020, [7] the law is still too recent to propagate throughout the industry as a standard. This could mean that universities prefer to follow only their legal obligations regarding privacy policy and do not see a benefit in implementing a DNSMPI link.

For instance, some universities that do not have a DNSMPI link may not see a reason to implement it if they do not sell consumers' personal information at all, as the link would be essentially useless. Therefore, it can be surmised that if the CPPA endorsed a statement option, such as "We Do Not Sell Your Personal Information," universities may be more inclined to voluntarily follow this standard, especially universities that do not intend to sell personal information.

4.2 RQ2

Our estimated DNSMPI compliance rate among American universities was 25% (7/28). Five of the 21 (23.8%) noncompliant universities were based or had campuses in California. These five were West Coast University, American Career College, California Northstate University, Pacific College of Health and Science, and Musicians Institute. All the universities above have undergraduate student populations under 3,000 students.

Sixteen universities (76.2%) have undergraduate student populations under 6,000 students. The five that did not were South College (app. 6,100), Monroe College (app. 7,000), ECPI University (app. 11,700), Columbia Southern University (app. 16,000), and Post University (app. 17,000). Notably, all these universities are in the eastern half of the US.

The results are available in A5/analysis/dnsmpi_analysis.txt in the repository, and the reviewed data is accessible through the links embedded in A5/data/dnsmpi_requirements.xlsx in the same repository.

4.2.1 Insight and Implications for RQ2

A compliance rate of 25% is simply too low to call the DNSMPI link a widely respected component of the CPPA. One potential explanation for this lack of compliance is the wording of "Do Not Sell My Personal Information." Universities may believe this wording will portray them negatively if readers believe they are selling consumers' personal information by default. Moreover, universities selling personal information may want to hide their activities by failing to include this link.

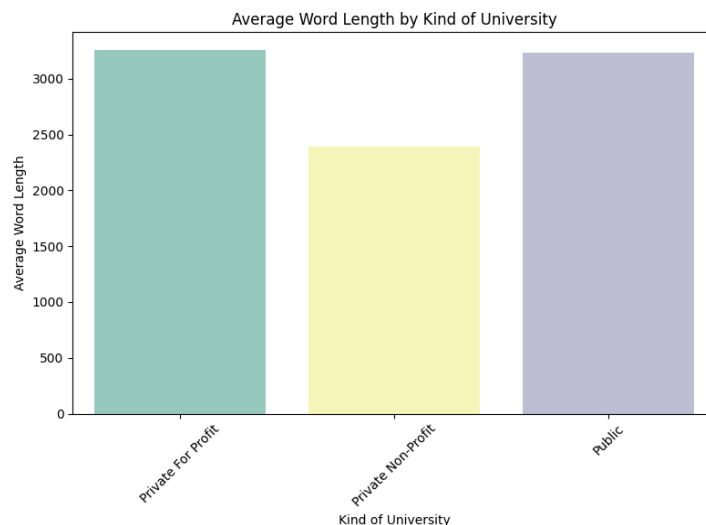
Some non-California universities may believe they can get away with not having a DNSMPI link because there is legal and jurisdictional ambiguity in how the CPPA would be enforced out of state. However, it is recommended that the state of California look more closely into the five California universities were identified as noncompliant with DNSMPI obligations.

Van Nortwick et al. found a DNSMPI adoption rate of 9% for the top 25,000 websites on the Tranco top list and 2,902 third-party A&A companies identified by Bashir et al. [7]. Given the high likelihood that these websites investigated in Van Nortwick et al. were obligated to include a DNSMPI link [7], it can be assumed that the DNSMPI compliance rate for the most popular websites is about 9%. This rate is significantly lower than the estimated compliance rate of 25%.

Some websites Van Nortwick et al. investigated were not American [7]. Therefore, one potential reason for the discrepancy in compliance rates is that this study observed only American universities. American universities may be more likely to comply with DNSMPI obligations than non-American universities because they are more inclined to do business with California.

4.3 RQ3

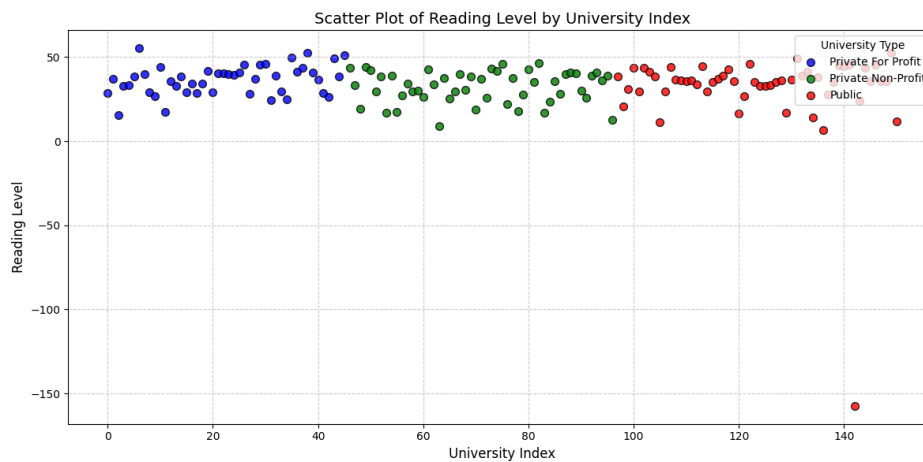
The data analysis revealed some notable differences and surprises in the word length, readability, and sentiment of privacy policies among public, private non-profit, and private for-profit universities. On average, private for-profit universities had the longest privacy policies, with an average word length of 3,252.93 words. Public universities followed closely with 3,235.41 words. Lastly, private non-profit universities had the shortest policies, averaging only 2,393.00 words.



Word Length Bar Graph- Shows the average word length among university type

Readability, as measured by the *Flesch Reading Ease Score*, showed that private for-profit universities produced the most readable policies with an average score of 36.57. Non-profit universities followed with a 32.35, and public universities had a 31.02. A score of 0-29 is considered “very confusing” and 90-100

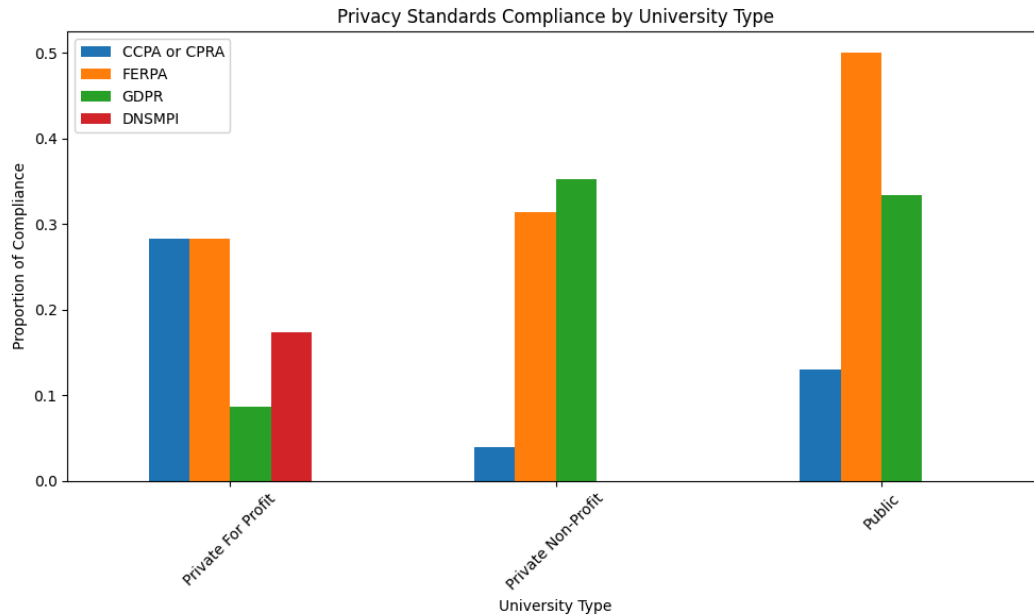
is considered “very easy”. This indicates that public universities had the most difficult-to-read policies while private-for-profits had the least difficult (comparatively speaking).



Readability Scatter Plot- Shows a fairly similar pattern of reading level among university type

In terms of sentiment, public universities exhibited the highest sentiment index at 0.07776, followed by private for-profits at 0.07449, and lastly private non-profits at 0.07288. The sentiment scale has -1 as “super negative” and 1 as “super positive”. The privacy policies seem to be overall neutral with no significant lean toward positive or negative sentiment.

Regarding the acknowledgment of key privacy laws, public universities demonstrated the highest acknowledgment rates of FERPA at 50% and the second highest GDPR at 33.33%. Meanwhile, private for-profits led in the acknowledgment of CCPA or CPRA at 28.26%. Private non-profits showed the highest GDPR acknowledgment at 35.29% but minimal acknowledgment to CCPA or CPRA at 3.92%. Private for-profits had the lowest GDPR acknowledgment at 8.7% and FERPA acknowledgment at 28.26%



Privacy law acknowledgment chart- shows acknowledgment averages among kinds of universities

4.3.1 Insight and Implications for RQ3

The results provide valuable insights into how universities communicate their privacy policies. For example, private for-profit universities' easier readability and relatively high word counts may reflect a strategic balance between legal thoroughness and user accessibility, possibly aiming to foster trust among potential students. Also, their high acknowledgment of U.S.-specific regulations like CCPA or CPRA and low acknowledgment of GDPR suggest a primarily domestic focus, potentially limiting the appeal to international audiences.

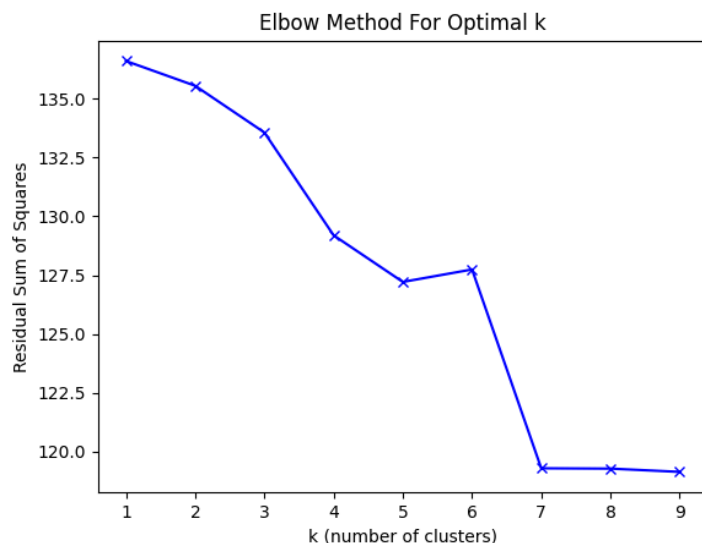
Initially, it was hypothesized that public universities, being government-funded and not having to rely heavily on personal data for profit, would have the easiest-to-read privacy policies. Their funding stability and limited dependence on traffic monetization were expected to result in the simplest and most accessible policies. However, these results contradict this assumption. In contrast, private for-profit universities, which might be expected to prioritize obfuscating legal safeguards due to their revenue-driven models, produced the most readable policies with an average readability score of 36.57. This suggests that for-profit institutions may be trying to potentially attract students by appearing transparent and trustworthy. It also can ensure compliance with regulations like CCPA or CPRA to avoid any restrictions that may impact their business model.

Private non-profit universities occupy a middle ground in readability, word length, and sentiment compared to public and private for-profit institutions. Their average readability score of 32.35 suggests their policies are more understandable than public universities (31.02) but less than private for-profits (36.57). With an average word length of 2,393, they have the shortest policies among the three groups, potentially reflecting an effort to balance detail and simplicity. Regarding sentiment, private non-profits scored 0.07288, closely aligning with the other groups but slightly less positive than public universities

(0.07776). This may indicate that non-profit institutions walk a fine line between regulation compliance and data collection. For example, this positioning may suggest that private non-profits aim to project a responsible image regarding data collection while minimizing risks of over disclosure.

Although including keywords such as GDPR, CCPA, CPRA, or FERPA (acronym and spelled out) do not correlate directly to adherence to the laws, it can still provide valuable insights. A key takeaway is from private for-profit universities' low acknowledgment of GDPR (8.7%). This shows that these universities may target domestic students more than international students. By not having to focus on international student acquisition, universities can continue practices that go against the GDPR since restrictions or penalties imposed by the EU most likely would have little to no impact on their business model. On the other hand, public and private non-profit universities have a much higher GDPR acknowledgment rate, both above 30%. This shows that these universities are making efforts to adhere to EU policy. By doing so, it could be assumed that these universities have a strong interest in international students and see them as an important part of their business model. Therefore, not adhering to EU laws could cause penalties or restrictions that limit potential business opportunities.

4.4 RQ4



The discovered optimal value of the number of clusters, k , was 7.

Cluster Number	Number of Universities	15 Significant Words
1	10	uw customers uc statement gdpr eu school products riverside communicating california diego platform highest san

2	43	california sensitive csu eea statement legitimate retention academic financial visitor code record european media employment
3	3	florida mail released sponsored computing atlantic instead accessed volume timing reach ideas recognizes designated official
4	7	illinois su isu statement tech western campus resources il academic pii graduate chicago just directory
5	27	college south claremont asu columbia office ferpa plaza district northwestern mexico www copyright georgia statement
6	10	usc apps sciences ads learn national hipaa arts oregon campus ad community app remarketing medicine
7	51	account device like activity ads apps settings california consumer search products learn categories location delete

The full results are available in the repository in A5/analysis/clusters.txt, A5/analysis/diction_by_cluster.txt, and A5/analysis/cluster_visualization.png.

4.4.1 Insight and Implications for RQ4

Clusters 1 and 2

Both clusters 1 and 2 have some similarities. Cluster 1 uses words like *gdpr* and *eu*, and cluster 2 has *european* and *eea*. This shows that both clusters want to attract students from and do business with the EU by following European privacy standards. Furthermore, both Cluster 1 and Cluster 2 mention California-specific terms like *uc*, *california*, *san diego*, and *riverside*, and *california* and *csu*, respectively. This makes sense since 5 out of 10 (50%) of the universities in Cluster 1 and 18 out of 43 (41.9%) of the universities in Cluster 2 are based in or have a campus in California.

Cluster 1 and Cluster 2 differ in how they view the student. Cluster 1 uses terms like *customers* and *products*, viewing the student as a customer purchasing a product. In contrast, Cluster 2 employs words such as *academic*, *visitor*, and *employment*, which relate to seeing the student more personally and less commercially. Moreover, Cluster 2 includes wording such as *sensitive*, *financial*, *record*, and *code*, all descriptors of the kinds of personal information the universities in Cluster 2 may encounter. Interestingly, half of the universities in Cluster 1 are public, whereas 21 out of 43 (48.8%) of the universities in Cluster 2 are private non-profit. Hence, of California universities that comply with EU privacy standards, the public universities tend to view the student as a consumer, whereas the private non-profit universities see the student more as a learner.

Cluster 3

The three universities in Cluster 3 were the three public universities from Florida. Hence, it is unsurprising that *florida* is the most significant word. Upon further analysis of the three universities'

privacy policies, it was found that they contain similar or the same wording and all refer, in one way or another, to Florida's Public Records Act.

The Public Records Act obligates state and governmental entities to allow public access to public records, such as email, unless certain information is specifically exempted [4]. Public universities in Florida must comply with this law as they are considered agencies of the state [6]. This is why wording relevant to the Public Records Act is used, including *mail*, *released*, *accessed*, *reached*, *official*, and *volume*.

Cluster 4

Cluster 4 contained several references to Illinois, including *Illinois*, *isu*, *il*, and *chicago*. The analysis reveals that this cluster contained four public universities from Illinois and DePaul University, also based in Illinois. One interesting word is *pii*, which stands for personally identifiable information. It is possible this word is significant because of Illinois's Personal Information Protection Act (PIPA), which requires public Illinois universities that collect PII to notify Illinois residents of a data breach [9].

Cluster 5

The most significant words in Cluster 5 were words shared among university names such as *south*, *college*, *claremont*, *columbia*, *northwestern*, *georgia*, and *mexico*. Interestingly, these privacy policies tended to be shorter than those in other clusters and contained less terminology related to personal privacy. Consequently, the relative frequency of words unrelated to personal privacy, like words shared by multiple university names, was inflated.

Cluster 6

Notably, *hippa*, *medicine*, *community*, and *sciences* are significant words for Cluster 6. 7 out of 10 (70%) universities in this cluster own hospitals and provide healthcare, and one is affiliated with a local hospital. Hence, most universities in this cluster structure privacy policies as they relate to, intersect with or complement medical laws.

Cluster 7

Cluster 7 contains wording like *account*, *device*, *activity*, *ads*, *apps*, *settings*, *consumer*, *search*, *products*, *categories*, *location*, and *delete*. These terms describe how a user interacts and uses their devices and views the student disproportionately as a consumer using a product. 23 of the 46 (50%) for-profit universities appeared in Cluster 7. 16 of the 23 (69.6%) offer online programs or are completely online. Moreover, 23 of the 51 (45.1%) universities were for-profit in this cluster. Because of this skew toward online and for-profit education in Cluster 7, it suggests that terms relating to online and consumer behavior appear significant in this sample group. This suggests that universities offering for-profit or online education may view students as consumers rather than learners.

Overall

Every cluster except Cluster 3 (85.7%) contained all three university classes (public, private for-profit, private non-profit). This implies that there are similarities among universities of different classes in their

privacy policies. However, a single class of universities comprised at least 45% of six (85.7%) of the seven clusters. In Cluster 3, a subclass of public universities is quite similar. This is the only case with an exact correlation between class and cluster. The observation was generally moderate, but there was not a full correlation between class and cluster. Therefore, the conclusion is that university classes are broad groups, internally diverse, and have some overlap with other university classes.

The cluster analysis identified four main factors that influence the legal priorities of privacy policies. First, the location or region in which a university lies. States other than California, like Florida and Illinois, have statewide privacy policy regulations that apply to public universities in those states. Second, is the desire of a university to attract students from or do business with the EU. If so, they often comply with the General Data Protection Regulation (GDPR). Third is how the university views students from a legal perspective. Private for-profit online universities are much more likely to see students as consumers than learners compared to the other two classes of universities. Lastly, it has been uncovered that universities with a major stake in providing a specific field of education, such as medicine, mention laws corresponding to that field regarding privacy in their privacy policies. For example, universities with a hospital system tend to mention the Health Insurance Portability and Accountability Act (HIPAA) in their privacy policies.

5 Limitations

5.1 RQ1 Limitations

A key limitation of the RQ1 analysis is the sample size of California universities. The study included data from 40 California universities. A larger sample size, with ideally at least 50 institutions, would have provided a more comprehensive basis for comparison. A more in-depth comparison would strengthen the reliability of the conclusions. Furthermore, the scope of the analysis did not account for potential variations in how universities interpret or prioritize compliance with privacy regulations such as the CPPA. It also does not consider factors such as institution size, available resources, or specific privacy practices. These factors could impact the adoption of DNSMPI links.

Additionally, the recent implementation of the CPPA in 2020 presents a temporal limitation. Organizations, including universities, may still be in the process of adjusting their practices to meet the requirements of the regulation. This limitation makes it challenging to assess long-term trends in DNSMPI link adoption. It is also difficult to determine whether these standards might achieve broader acceptance in the future.

5.2 RQ2 Limitations

A primary limitation of the RQ2 analysis is the relatively small sample size of universities required to implement DNSMPI links. With only 28 institutions included in this study, the findings may not fully capture compliance behavior trends across the population. A larger sample size, ideally including at least

50 universities subject to DNSMPI obligations, would provide a stronger foundation for analysis. This would enhance the generalization of the results.

Additionally, the study does not account for potential differences in legal interpretations or enforcement strategies by the CPPA, particularly concerning out-of-state universities. Legal and jurisdictional ambiguities may influence compliance behaviors. However, this area has not been explored.

Furthermore, the analysis does not consider institutional factors such as budget restraints, administrative priorities, or reputational concerns. These concerns may impact a university's decision to comply with DNSMPI requirements. Finally, the study's use of headless browsers outside of the state of California could be missing compliance within the state. The current research method does not consider the possibility of websites limiting DNSMPI link visibility to certain states.

5.3 RQ3 Limitations

The method used to calculate word count and readability is a limitation of the RQ3 analysis. Determining word length followed a simple methodology of splitting the string into a list and then returning the length of the list. For better accuracy, the study could have used a specialized library that determines word length or could have implemented a more advanced algorithm. Clarifying the study's assumption regarding word count and ambiguity is also important. While this research hypothesizes that higher word counts in privacy policies may indicate efforts to obfuscate or create ambiguity, this is not definitive. A higher word count could reflect a university's attempt to provide detailed explanations, transparency, or full coverage of their policies. Therefore, without additional context and analysis, word count alone cannot serve as a reliable indicator of ambiguity or obfuscation.

Additionally, checking for keywords like GDPR, CCPA, CPRA, or FERPA in privacy policies does not guarantee actual compliance with the regulations. The study assumes that keyword inclusion shows some level of adherence. However, it does not determine if the laws are being followed when collecting, storing, or processing user data. The study cannot definitively link keyword mentions to legal compliance without analyzing the underlying practices.

Another limitation involves potential bias in interpreting readability and sentiment scores. These metrics follow algorithms that may not fully capture the user experience or how the individual users perceive and understand privacy policies.

5.4 RQ4 Limitations

One limitation of the RQ4 analysis lies in the inherent randomness of the k-means clustering process. The algorithm randomly initializes cluster centroids, resulting in different clusters, especially with limited iterations. Although the study mitigates this issue by running the k-means process 100,000 times to reduce randomness, the possibility of slightly different clustering outcomes still exists.

Another limitation stems from the subjectivity in determining the optimal number of clusters, k . Identifying the "elbow" point in the graph of increasing k values and the corresponding sum of squared distances is not an exact science and can vary based on interpretation. This subjectivity introduces a risk that the selected k may not represent the true optimal value.

Finally, the analysis is limited by the presence of university names and common identifiers, such as "college." These terms comprise a significant portion of the clusters, possibly skewing the analysis results.

6 Future Direction

6.1 Dataset

The study dataset consists of 151 educational institutions, where 54 are public, 51 are private non-profit, and 46 are private for-profit. Future research should create a larger dataset and evenly distribute how many public, private non-profit, and private for-profit are to be analyzed. This would allow for a richer and more accurate analysis as the dataset would have more variety and have even proportions of type of university.

6.2 Data Analysis

For universities that mentioned privacy policies such as CCPA, GDPR, or FERPA, future research would like to investigate whether universities comply with the laws mentioned. This study only checks acknowledgment of policies but not actual compliance.

This study uses the *Flesch Reading Ease Formula* to determine readability. A formula cannot perfectly match how a reader perceives and comprehends the policy data. Therefore, future research could include a group of diverse readers to give deeper insight into perception and comprehension. Insights from real readers would provide a better understanding of whether privacy policies are accessible and understandable to their audience of students, parents, and faculty.

6.3 Broader Geographic Analysis

This study focused on three categories of educational institutions in the United States. Future research can expand the study into the EU (European Union) and compare whether the trends are the same. Do the three categories in the USA match the trends of the three categories in the EU? For example, this study has findings that indicate for-profit universities do not target international students. Is that true for the EU as well?

7 Conclusion

This study has narrowed the gap in existing research between privacy policies and universities and has made some useful conclusions that the study hopes will encourage further research in this topic.

7.1 RQ1 Conclusion

The study sought to understand the cultural or legal prevalence of the CPPA/CPRA on voluntary DNSMPI link adoption. To do so, the study computed the proportion of privacy policies that voluntarily adopted DNSMPI links inside California (2.5%) and outside California (0%). Due to the extremely low voluntary adoption rates, the study found no evidence of a local industry standard for DNSMPI link adoption. The study concludes that most DNSMPI-exempt universities do not see a benefit in implementing a DNSMPI link.

7.2 RQ2 Conclusion

The study endeavored to find the DNSMPI compliance rate among American universities and locate any noncompliant universities. The study identified which universities were legally obligated to have DNSMPI links, identified the ones that did not, and computed an overall compliance rate of 25%. The study identified five noncompliant universities based in California and found that the noncompliant universities with the largest student populations tended to lie in the eastern half of the US. This low compliance rate may be due to California's legal and jurisdictional ambiguity in out-of-state enforcement of the CPPA and CPRA.

7.3 RQ3 Conclusion

This study analyzed privacy policies of American institutions to uncover differences in transparency, readability, and law acknowledgment across public, private non-profit, and private for-profit institutions. Private for-profits had the longest but most readable policies. Meanwhile, public universities were the least readable despite initial belief. Findings highlighted differing priorities among universities. For-profits focused on domestic laws like CCPA, which may highlight a focus on domestic students. On the other hand, public and non-profits showed a stronger focus on GDPR, potentially reflecting international student interest. However, word count methods, readability metrics, and compliance assessment limitations call for larger datasets, user feedback, and deeper compliance analysis for future research.

7.4 RQ4 Conclusion

The study examined the strength of the correlation between university class (public, private for-profit, private non-profit) and significant wording and what the significant wording in privacy policies reveals about American universities' legal priorities. The study utilized k-means clustering and identified seven distinct clusters to do this. It was found that university classes have broad, diverse groups of privacy policies that overlap with other classes' privacy policies in certain cases.

Four main legal factors have been noted in which privacy policies share similar wording. The first is location, as states like California, Florida, and Illinois all have statewide privacy regulations that apply to their public universities. The second is compliance with EU privacy regulations such as GDPR. Third is the tendency to view the student as a consumer or a learner. The last factor is the education offered, especially concerning medicine and the intersection of privacy law with HIPPA.

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